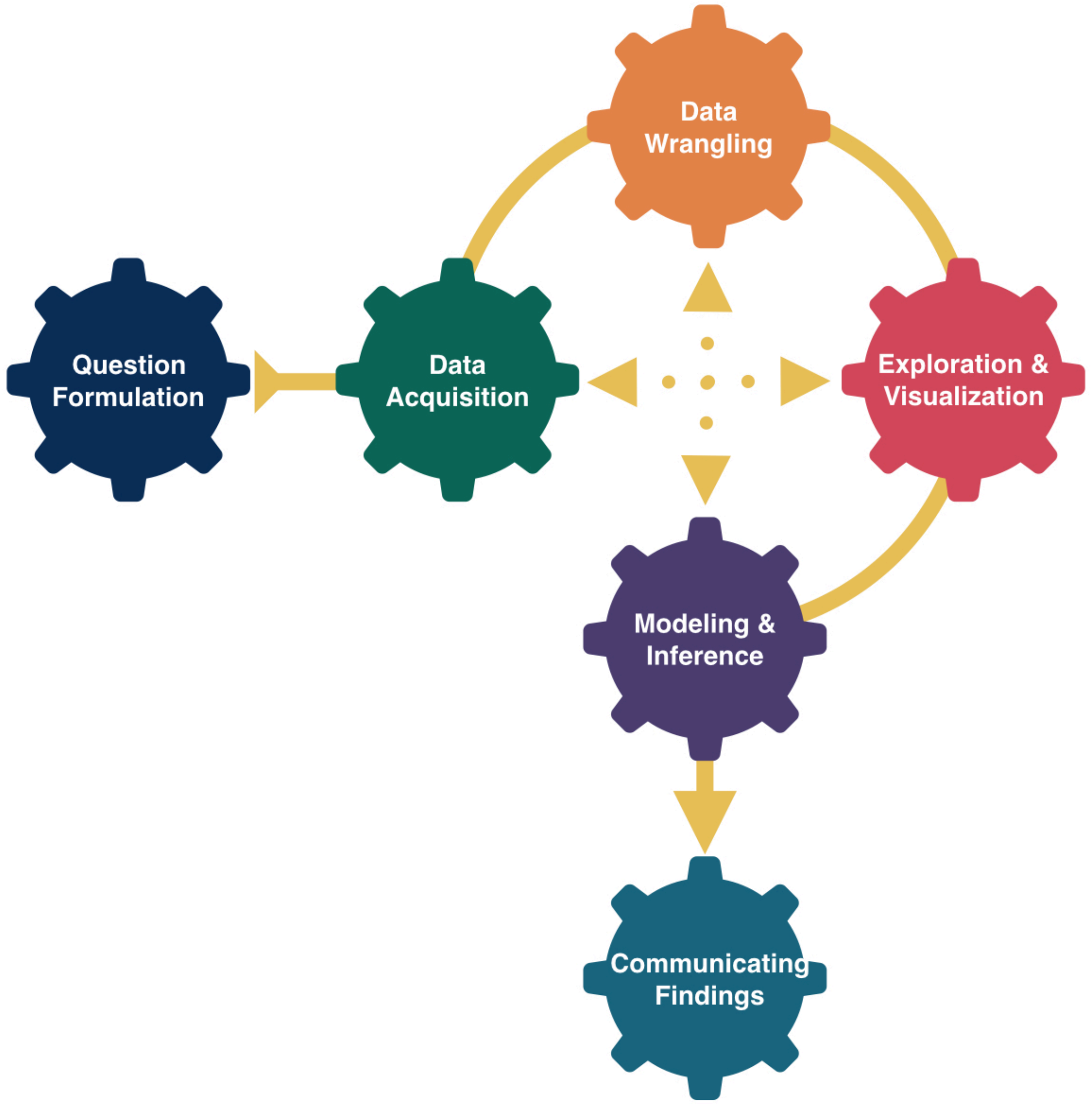



# Model Guidance

Kelly McConville

DATA 100

Week 7 | Spring 2026

# Announcements

- Remember: Spring Break next week
- Friday: Optional lab, no quiz. But will receive a problem set (due on the Thursday after break).

## Goals for Today

- Modeling guidance
- A bit of conceptual review

# Which Are You?

Data Visualizer

Data Wrangler

Model Builder

# Linear Regression

Model Form:

$$y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

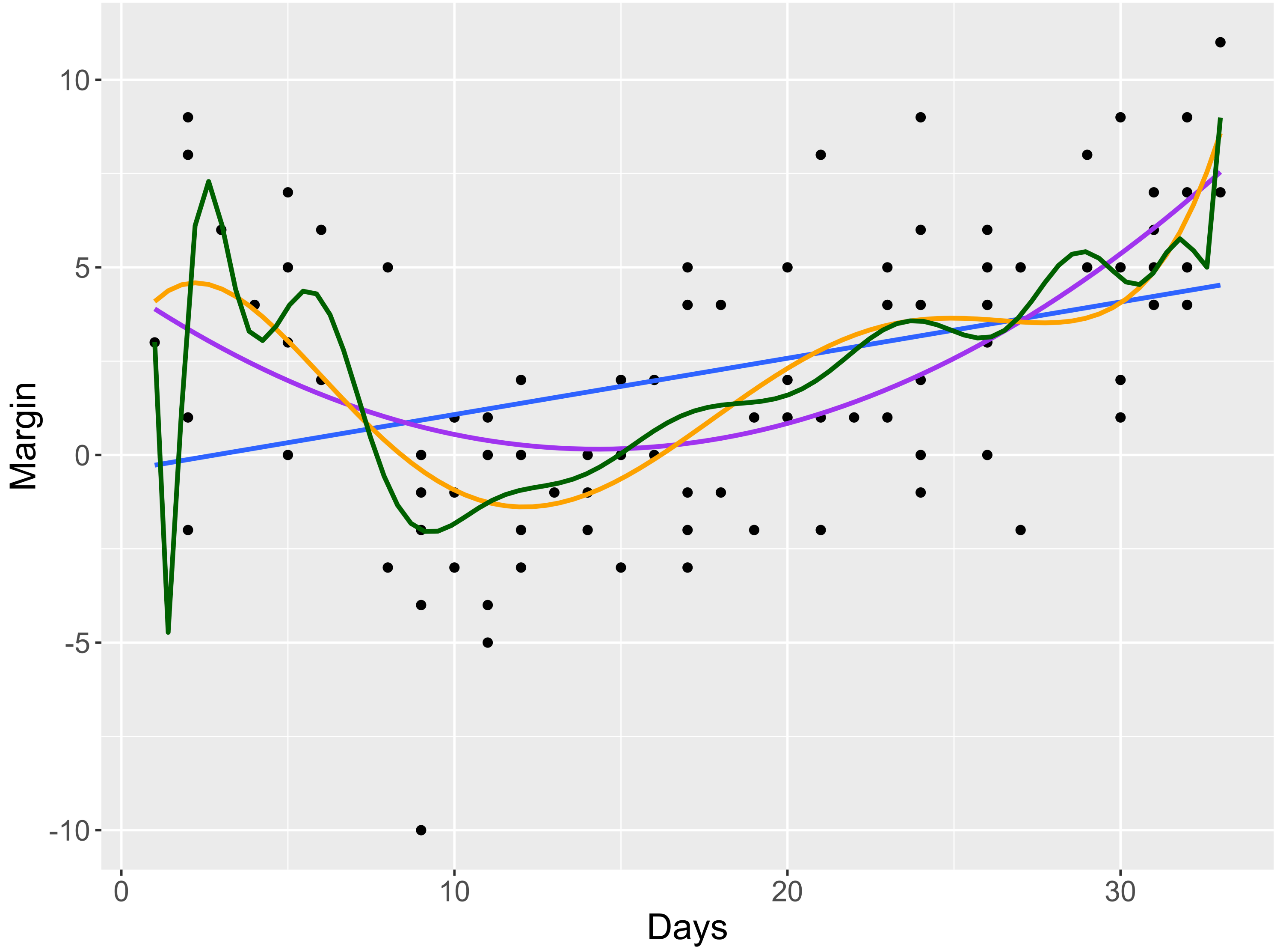
Linear regression is a flexible class of models that allow for:

- Both quantitative and categorical explanatory variables.
- Multiple explanatory variables.
- Curved relationships between the response variable and the explanatory variable.
- BUT the response variable is quantitative.

# Model Building Guidance

What degree of polynomial should I include in my model?

Returning the 2008 Election Example:




**Guiding Principle**: Capture the general trend, not the noise.

$$y = f(x) + \epsilon$$
$$y = \text{TREND} + \text{NOISE}$$

# Model Building Guidance

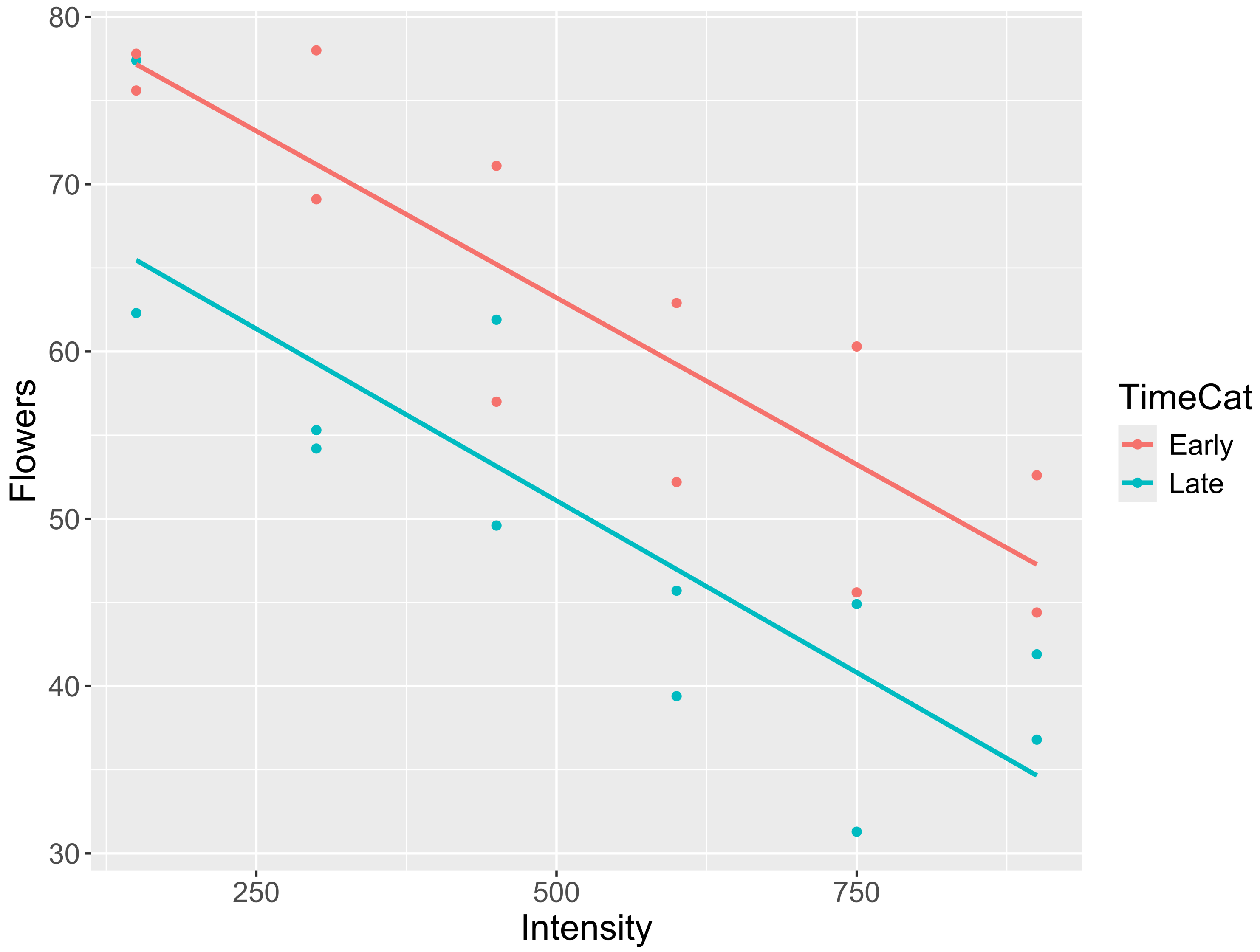
Shouldn't we always include the interaction term?

Guiding Principle: Occam's Razor for Modeling

> "All other things being equal, simpler models are to be preferred over complex ones." – ModernDive

Guiding Principle: Consider your modeling goals.

- The equal slopes model allows us to control for the intensity of the light and then see the impact of being in the early or late timing groups on the number of flowers.




- Later in the course will learn statistical procedures for determining whether or not a particular term should be included in the model.

# What if I want to include more than 2 explanatory variables??

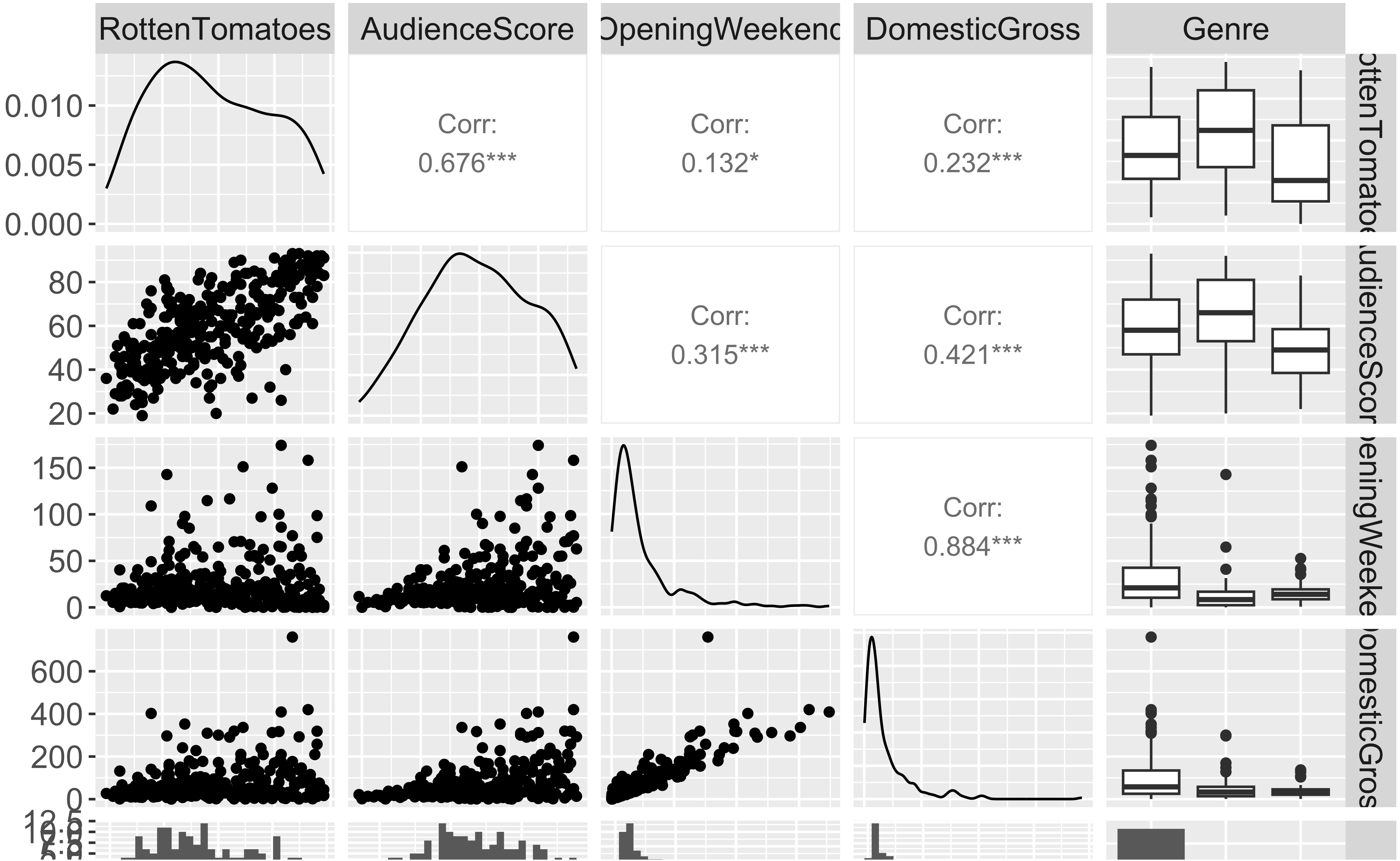
# Model Building Guidance

We often have several potential explanatory variables. How do we determine which to include in the model and in what form?

**Guiding Principle**: Include explanatory variables that attempt to explain **different** aspects of the variation in the response variable.

```
Rows: 313
Columns: 16
$ Movie            <chr> "Spider-Man 3", "Transformers", "Pirates of the Carib…
$ LeadStudio       <chr> "Sony", "Paramount", "Disney", "Warner Bros", "Warner…
$ RottenTomatoes   <dbl> 61, 57, 45, 60, 20, 79, 35, 28, 41, 71, 95, 42, 18, 2…
$ AudienceScore    <dbl> 54, 89, 74, 90, 68, 86, 55, 56, 81, 52, 84, 55, 70, 6…
$ Story            <chr> "Metamorphosis", "Monster Force", "Rescue", "Sacrific…
$ Genre            <chr> "Action", "Action", "Action", "Action", "Action", "Ac…
$ TheatersOpenWeek <dbl> 4252, 4011, 4362, 3103, 3778, 3408, 3959, 3619, 2911,…
$ OpeningWeekend   <dbl> 151.1, 70.5, 114.7, 70.9, 49.1, 33.4, 58.0, 45.3, 19.…
$ BOAvgOpenWeekend <dbl> 35540, 17577, 26302, 22844, 12996, 9791, 14663, 12541…
$ DomesticGross    <dbl> 336.53, 319.25, 309.42, 210.61, 140.13, 134.53, 131.9…
$ ForeignGross     <dbl> 554.34, 390.46, 654.00, 245.45, 117.90, 249.00, 157.1…
$ WorldGross       <dbl> 890.87, 709.71, 963.42, 456.07, 258.02, 383.53, 289.0…
```

```
library(GGally)
movies2 %>%
  select(RottenTomatoes, AudienceScore, OpeningWeekend,
         DomesticGross, Genre) %>%
  ggpairs()
```


RottenTomatoes
AudienceScore
OpeningWeekend
DomesticGross
Genre
Corr:
0.676***
Corr:
0.132*
Corr:
0.232***
Corr:
0.315***
Corr:
0.421***
Corr:
0.884***
0.010
0.005
0.000
80
60
40
20
150
100
50
0
600
400
200
0

# Model Building Guidance

We often have several potential explanatory variables. How do we determine which to include in the model and in what form?

**Guiding Principle**: Include explanatory variables that attempt to explain **different** aspects of the variation in the response variable.

```r
mod_movies <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)
get_regression_table(mod_movies, print = TRUE)
```

| term | estimate | std_error | statistic | p_value | lower_ci | upper_ci |
|---|---|---|---|---|---|---|
| intercept | -12.472 | 4.083 | -3.055 | 0.002 | -20.506 | -4.439 |
| AudienceScore | 0.975 | 0.072 | 13.590 | 0.000 | 0.834 | 1.117 |
| Genre: Drama | 6.117 | 2.644 | 2.314 | 0.021 | 0.916 | 11.319 |
| Genre: Horror | 2.058 | 3.141 | 0.655 | 0.513 | -4.121 | 8.238 |
| DomesticGross | -0.006 | 0.015 | -0.431 | 0.667 | -0.035 | 0.023 |

# Model Building Guidance

Suppose I built 2 different models. **Which is better?**

```r
mod1 <- lm(RottenTomatoes ~ AudienceScore, data = movies2)
mod2 <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)
```

- Big question! Take a modeling course to learn systematic model selection techniques.
- We will explore one approach. (But there are many possible approaches!)

# Comparing Models

Suppose I built 2 different models. **Which is better?**

- Pick the best model based on some measure of quality.

**Measure of quality**: $R^2$ (Coefficient of Determination)

$$\begin{aligned} R^2 &= \text{Percent of total variation in y explained by the model} \\ &= 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2} \end{aligned}$$

**Strategy**: Compute the $R^2$ value for each model and pick the one with the highest $R^2$.

# Comparing Models with $R^2$

**Strategy**: Compute the $R^2$ value for each model and pick the one with the highest $R^2$.

```r
mod1 <- lm(RottenTomatoes ~ AudienceScore + Genre, data = movies2)
mod2 <- lm(RottenTomatoes ~ AudienceScore + Genre + DomesticGross, data = movies2)

get_regression_summaries(mod1)
```

```
# A tibble: 1 × 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
      <dbl>         <dbl> <dbl> <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl>
1     0.469         0.464  353.  18.8  18.9      90.9       0     3   313
```

```r
get_regression_summaries(mod2)
```

```
# A tibble: 1 × 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
      <dbl>         <dbl> <dbl> <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl>
1     0.469         0.462  353.  18.8  18.9      68.0       0     4   313
```

**Strategy**: Compute the $R^2$ value for each model and pick the one with the highest $R^2$.

```r
get_regression_summaries(mod1)
```

```
# A tibble: 1 × 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
      <dbl>         <dbl> <dbl> <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl>
1     0.469         0.464  353.  18.8  18.9      90.9       0     3   313
```

```r
get_regression_summaries(mod2)
```

```
# A tibble: 1 × 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
      <dbl>         <dbl> <dbl> <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl>
1     0.469         0.462  353.  18.8  18.9      68.0       0     4   313
```

**Problem:** As we add predictors, the $R^2$ value will only increase.

**Guiding Principle**: Occam's Razor for Modeling

> "All other things being equal, simpler models are to be preferred over complex ones." – ModernDive

# Comparing Models with the Adjusted $R^2$

**New Measure of quality**: Adjusted $R^2$ (Coefficient of Determination)

$$\text{adj}R^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}\left(\frac{n-1}{n-p-1}\right)$$

where $p$ is the number of explanatory variables in the model.

- Now we will penalize larger models.
- **Strategy**: Compute the adjusted $R^2$ value for each model and pick the one with the highest adjusted $R^2$.

**Strategy**: Compute the adjusted $R^2$ value for each model and pick the one with the highest adjusted $R^2$.

```r
get_regression_summaries(mod1)
```

```
# A tibble: 1 × 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
      <dbl>         <dbl> <dbl> <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl>
1     0.469         0.464  353.  18.8  18.9      90.9       0     3   313
```

```r
get_regression_summaries(mod2)
```

```
# A tibble: 1 × 9
  r_squared adj_r_squared   mse  rmse sigma statistic p_value    df  nobs
      <dbl>         <dbl> <dbl> <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl>
1     0.469         0.462  353.  18.8  18.9      68.0       0     4   313
```

# Model Building Guidance

We often have several potential explanatory variables. How do we determine which to include in the model and in what form?

Guiding Principle: Use your modeling motivation to determine how much you weigh interpretability versus prediction accuracy when choosing the model.

# Model Building

- We will come back to methods for model selection.
- Key ideas:
  - Determining the **response** variable and the potential **explanatory** variable(s)
  - Writing out the **model form** and understanding what the terms represent **in context**
  - **Building** and **visualizing** linear regression models in `R`
  - **Comparing** different potential models
- After break: Look at models before linear regression!

# Shift Gears: Conceptual Review

# One can conclude a causal relationship between two variables if the study contains random sampling.

TRUE or FALSE

(1) or (2)

# The larger the value of correlation coefficient, the stronger the linear relationship between two variables.

TRUE or FALSE

(1) or (2)

# The IQR is a measure of the variability of the middle 50% of the data.

TRUE or FALSE

(1) or (2)

# A residual can only be non-negative.

TRUE or FALSE

(1) or (2)

# A non-random sample of 10,000 individuals is better than a random sample of 1000.

TRUE or FALSE

(1) or (2)

# Bar graphs and histograms are the same graph since they both contain bars.

TRUE or FALSE

(1) or (2)

# When using a linear regression model to make predictions, you can only make predictions at the observed values for your explanatory variables.

TRUE or FALSE

(1) or (2)

# Name Those Functions

## Unwrangled Data

```
# A tibble: 5 × 4
  species island bill_length_mm sex   
  <fct>   <fct>           <dbl> <fct> 
1 Gentoo  Biscoe           45.2 female
2 Gentoo  Biscoe           50.5 male  
3 Adelie  Biscoe           38.8 male  
4 Gentoo  Biscoe           50.5 male  
5 Gentoo  Biscoe           45.3 female
```

```
full
```

```
# A tibble: 6 × 4
  species island bill_length_mm sex   
  <fct>   <fct>           <dbl> <fct> 
1 Gentoo  Biscoe           45.2 female
2 Gentoo  Biscoe           50.5 male  
3 Adelie  Biscoe           38.8 male  
4 Gentoo  Biscoe           50.5 male  
5 Adelie  Dream            37.5 <NA>  
6 Gentoo  Biscoe           45.3 female
```

## Wrangled Data

```
wrangled
```

```
# A tibble: 5 × 4
  species island bill_length_mm sex   
  <fct>   <fct>           <dbl> <fct> 
1 Gentoo  Biscoe           45.2 female
2 Gentoo  Biscoe           50.5 male  
3 Adelie  Biscoe           38.8 male  
4 Gentoo  Biscoe           50.5 male  
5 Gentoo  Biscoe           45.3 female
```

What `dplyr` function(s) did I use to wrangle these data?

# Name Those Functions

## Unwrangled Data

```
full
```

```
# A tibble: 6 × 4
  species island bill_length_mm sex
  <fct>   <fct>           <dbl> <fct>
1 Gentoo  Biscoe           45.2 female
2 Gentoo  Biscoe           50.5 male
3 Adelie  Biscoe           38.8 male
4 Gentoo  Biscoe           50.5 male
5 Adelie  Dream            37.5 <NA>
6 Gentoo  Biscoe           45.3 female
```

## Wrangled Data

```
wrangled
```

```
# A tibble: 4 × 5
# Groups:   island [2]
  species island sex        n     p
  <fct>   <fct>  <fct>  <int> <dbl>
1 Adelie  Biscoe male       1   0.2
2 Adelie  Dream  <NA>       1   1
3 Gentoo  Biscoe female     2   0.4
4 Gentoo  Biscoe male       2   0.4
```

What `dplyr` function(s) did I use to wrangle these data?

# Name Those Functions

## Unwrangled Data

```
full
```

```
# A tibble: 6 × 4
  species island bill_length_mm sex
  <fct>   <fct>           <dbl> <fct>
1 Gentoo  Biscoe           45.2 female
2 Gentoo  Biscoe           50.5 male
3 Adelie  Biscoe           38.8 male
4 Gentoo  Biscoe           50.5 male
5 Adelie  Dream            37.5 <NA>
6 Gentoo  Biscoe           45.3 female
```

## Wrangled Data

```
wrangled
```

```
# A tibble: 2 × 2
  species island
  <fct>   <fct>
1 Adelie  Biscoe
2 Adelie  Dream
```

What `dplyr` function(s) did I use to wrangle these data?

# Name Those Functions

## Unwrangled Data

```
full
```

```
# A tibble: 6 × 4
  species island bill_length_mm sex
  <fct>   <fct>           <dbl> <fct>
1 Gentoo  Biscoe           45.2 female
2 Gentoo  Biscoe           50.5 male
3 Adelie  Biscoe           38.8 male
4 Gentoo  Biscoe           50.5 male
5 Adelie  Dream            37.5 <NA>
6 Gentoo  Biscoe           45.3 female
```

## Wrangled Data

```
wrangled
```

```
# A tibble: 1 × 1
  mean_bill
      <dbl>
1      44.6
```

What `dplyr` function(s) did I use to wrangle these data?

# Reminders:

- Remember: Spring Break next week
- Friday: Optional lab, no quiz. But will receive a problem set (due on the Thursday after break).